

GLOBAL AI ADOPTION IN EDUCATION

A COMPREHENSIVE GLOBAL PERFORMANCE INTELLIGENCE REPORT

A Data-Driven Analysis of Student, Teacher, and School-Level AI Adoption Across Global Education Systems (2015–2026)

Field	Detail
Student Name	Shubham Vitthal Patil
Program	Data Analytics with Tableau
Project Type	Individual
Tools	Tableau Desktop / Tableau Public
Phase	Project Initialization & Planning
Dataset	Global_AI_in_Education.csv (2015–2026, 10 countries, 6 regions, 1,360 records)
Date	03 July 2026

1. Define Problem Statement

Artificial intelligence is being adopted across classrooms, teaching staff, and school systems at very different speeds around the world. While student-facing tools such as ChatGPT, Google Gemini, Microsoft Copilot, and Khanmigo are increasingly common, education stakeholders still lack a clear, comparative view of how adoption differs across countries, regions, urban versus rural settings, and over time. Policymakers, school administrators, and edtech vendors need a structured way to see where AI adoption is accelerating, where it is lagging, and what factors — such as internet penetration, government policy, and curriculum integration — are driving the gap.

This project addresses the problem by analysing a global AI-in-education dataset spanning 2015 to 2026 using Tableau to uncover meaningful patterns and trends. The dataset includes key attributes such as Year, Month, Country, Region, Student AI Usage %, Teacher AI Usage %, School AI Adoption %, Average Daily AI Usage (minutes), Top AI Tool, Internet Penetration %, Education Index, Government AI Policy status, Urban/Rural AI Usage %, AI-in-Curriculum status, and Gender Gap in AI Usage %. By leveraging interactive visualizations, the project enables a deeper understanding of how these factors shape AI adoption outcomes across the global education landscape.

Through this analysis, the project identifies critical insights such as the fastest-adopting countries and regions, the equity gap between urban and rural AI usage, the relationship between government policy and curriculum integration, and the year-over-year acceleration in student and teacher AI usage. These findings help highlight both standout success stories and persistent gaps in global AI-education readiness.

Core Business Question

How can data-driven BI insights from global AI-in-education data empower policymakers, school leaders, and edtech stakeholders to make informed decisions about digital equity, curriculum policy, teacher enablement, and AI-tool investment?

Key Problem Dimensions

- AI usage distribution across students and teachers, by country, region, and year
- Identification of high-adoption countries/regions and leading AI tools in education
- Urban vs. rural AI usage gap and its relationship to internet penetration and education index
- Government AI policy status and its correlation with AI-in-curriculum adoption
- Trend analysis to detect year-over-year growth in AI usage and the gender gap in adoption

2. Project Planning

Work Breakdown Structure (Epics)

Epic / Task	Description
Data Collection & Extraction	Gather the global AI-in-education dataset covering Year, Month, Country, Region, Student/Teacher AI Usage, School AI Adoption, Daily Usage Minutes, Top AI Tool, Internet Penetration, Education Index, Government Policy, Urban/Rural Usage, Curriculum Status, and Gender Gap
Data Preparation	Clean, normalize, and structure the raw CSV data; handle missing Government AI Policy values, duplicates, and type inconsistencies for Tableau readiness
Data Visualizations	Build individual charts — adoption trend lines, regional heatmaps, country comparisons, urban/rural and gender-gap breakdowns, and top-tool usage charts
Dashboard	Assemble an interactive multi-view dashboard linking all visualizations with filters (year, country, region, policy status) and cross-highlighting
Story	Create a Tableau Story with annotated narrative slides guiding stakeholders through key AI-in-education adoption findings
Performance Testing	Validate dashboard load time, filter responsiveness, and cross-visualization accuracy
Web Integration	Publish to Tableau Public; integrate demo link and GitHub repository link into project documentation
Project Demonstration & Documentation	Record a walkthrough video, finalize README, and prepare presentation materials

Tools & Technology

- Tableau Desktop / Tableau Public — primary BI and visualization platform
- Microsoft Excel / CSV — dataset format for ingestion (Global_AI_in_Education.csv)
- GitHub — version control and project documentation repository

3. Proposed Solution

The solution leverages Tableau’s interactive BI capabilities to produce a data-driven platform that delivers targeted insights drawn from the global AI-in-education dataset. The analysis focuses on uncovering patterns in adoption trends, regional and country-level disparities, the urban-rural digital divide, and the policy-to-classroom pipeline.

Insight-Specific Solutions

- **Geographic & Regional Analysis** — country and region-wise comparison of student, teacher, and school AI adoption rates to identify leading and lagging markets.
- **Policy & Curriculum Insights** — analysis of government AI policy status (None / Draft / Active) against AI-in-curriculum adoption to reveal how policy translates into classroom practice.
- **Equity & Access Analysis** — urban vs. rural AI usage gap and gender gap in AI usage, cross-referenced with internet penetration and education index, to surface digital equity concerns.
- **Adoption & Tool Trend Analysis** — year-over-year growth in AI usage (2015–2026) and comparison of leading AI tools (ChatGPT, Google Gemini, Microsoft Copilot, Khanmigo) by region.

Project Deliverables

Deliverable	Description
Interactive Dashboard	Multi-tab Tableau dashboard with filters covering year, country/region, policy status, urban/rural, and AI tool dimensions
Tableau Story	Annotated story arc walking through key findings on adoption trends, policy impact, and equity gaps with data-backed recommendations
GitHub Repository	Full project source including dataset, workbook (.twbx), and documentation README
Demo Recording	Walkthrough video demonstrating dashboard interactions and key insights for mentor evaluation

Success Criteria

- Dashboard clearly addresses all key analytical dimensions with dedicated filter views
- Adoption trends, policy/curriculum relationships, urban-rural and gender-gap insights are accurate and reproducible from the source dataset

- Story narrative is coherent and communicates actionable policy and edtech recommendations
- Published Tableau Public workbook is accessible and loads within acceptable time
- GitHub repository is documented with README, dataset reference, and workbook file